xphd: how to run

Exciton–phonon matrix elements, linewidths, symmetry, transport and photoemission from Quantum ESPRESSO, LetzElPhC and yambo

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This is the only document. It lists what the upstream codes must produce, every command in the order it is run, what each needs and writes, and how to tell that its output is right. Every option of every command is printed by `xphd <command> --help` and collected in the reference at the end. Every run writes a report next to its output, a PDF on Windows and a text file on Linux (Sec. 6).

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1. What xphd needs and what it gives

Needs a yambo `SAVE` (from `p2y` on the nscf); the BSE at every irreducible $\mathbf{Q}$ with full diagonalisation (`ndb.BS_diago_Q*`); LetzElPhC's `ndb.elph` built on *the same* `SAVE`, over the BSE band window; `matdyn.x` frequencies (and eigenvectors) on a fine q mesh; optionally the GW `ndb.QP` and `ndb.dipoles`.

Gives exciton–phonon matrix elements $G(\mathbf{Q}, \mathbf{q})$ on the full zone, one archive per $\mathbf{Q}$; broadening-free linewidths, self-energies and absorption lineshapes; the mirror (σ_h) parity and irreducible representations of excitons, bands and phonons, and the selection rules they imply; exciton dipoles and chirality; real-time Boltzmann transport of the exciton populations; exciton photoemission maps (thermal or time-resolved).

2. Install

```
cd xphd
pip install -e ".[plot,yambo]"      # numpy, matplotlib, netCDF4
pip install <path to yambopy>        # from source; needed by every command that reads yambo or LetzElPhC
pip install -e ".[hpc]"              # clusters: mpi4py (built against the cluster MPI), scipy, threadpoolctl
xphd --help                          # every command
xphd <command> --help                # its options
python -m pytest                     # synthetic data; no yambo databases needed
```

Only `numpy` is required to import the package; the other libraries are imported by the commands that need them. `scipy` is needed by `bte`. Paths in angle brackets are placeholders.

3. The workflow at a glance

Stage	Code	Output
A. Ground state	`pw.x` scf, nscf; `p2y`	`SAVE`
B. Phonons	`ph.x`; `q2r.x`; `matdyn.x`	dyn, dvscf; force constants; fine-mesh frequencies
C. Excitons	yambo BSE (GW optional)	`ndb.BS_diago_Q1` … at every irreducible $\mathbf{Q}$
C′. El–ph	LetzElPhC	`ndb.elph` (same `SAVE` as C)
D. Archives	`xphd dmats`, `generate`, `check-archive`, `verify-archives`	`Dmats.npy`, `GI_ExcPh_Q0001.npz` …
E. Fine phonons	`xphd matdyn-split`, `matdyn-merge`, `modes-check`, `matdyn`, `mode-labels`	`hw_fine.npy`, `mode_labels.npy`
F. Widths	`xphd linewidth`, `sweep`, `unfold`, `selfenergy`, `lineshape`	widths, zone maps, $\Sigma(\omega)$, ε_2
G. Symmetry	`xphd parity`, `band-parity`, `irreps`, `selection-rule`	parity maps, labels, allowed/forbidden split
H. Optics	`xphd dipoles`, `chirality`	`exc_dipoles.npy`, helicity
I. Dynamics	`xphd bte`, `arpes`	snapshots; photoemission maps

D needs A–C′; E needs B; F needs D and E; G needs A, C and D; I needs D and E (and G for `bte --window`).

4. Upstream settings

xphd trusts its inputs; these must hold.

4.1 Quantum ESPRESSO

- scf, phonons and nscf in separate directories (the nscf overwrites `<prefix>.save`).
- q-grid = k-grid ($N \times N \times 1$), so every $\mathbf{k} + \mathbf{q}$ lies on the mesh. Set `fildvscf` in `ph.x`.
- A `ph.x` restart with `recover=.true.` trusts the records in `_ph0/<prefix>.phsave`: remove the `dynmat.<iq>.*.xml` of any $\mathbf{q}$ to be recomputed first, or its dvscf is left empty. Check that every dvscf has the same size.
- Spin–orbit: `noncolin=.true.`, `lspinorb=.true.`, fully relativistic pseudopotentials; `ph.x` then writes spinor dvscf.

4.2 q2r.x and matdyn.x

```
q2r.in                              matdyn.in.tmpl  (namelist only; the q-list is appended per chunk)
&input                              &input
  fildyn  = '<prefix>.dyn'            asr = 'crystal', flfrc = '<prefix>.fc'
  zasr    = 'crystal'                 flfrq = 'FLFRQ', flvec = 'FLVEC'
  flfrc   = '<prefix>.fc'             loto_2d = .true., q_in_cryst_coord = .true.
  loto_2d = .true.                    q_in_band_form = .false., nosym = .true., dos = .false.
/                                   /
```

`loto_2d=.true.` in both (the 3D non-analytic term adds a spurious LO splitting at Γ). The fine mesh is $N_f = N \times r$ with r the refinement used later (e.g. $18 \times 20 = 360$).

4.3 LetzElPhC

```
nqpool      = 4               # nqpool x nkpool = the MPI ranks
nkpool      = 12
start_bnd   = <first BSE band>
end_bnd     = <last BSE band>
save_dir    = <SAVE>          # the SAVE the BSE reads
ph_save_dir = ph.save         # the default is ph_save; misspelt keys are ignored silently
kernel      = dfpt
convention  = yambo
```

`ph.save` gathers `dyn0..N`, `dvscf1..N`, `patterns.N.xml`, `tensors.xml`, `data-file-schema.xml` and the pseudopotentials (LetzElPhC's `lelphc -pp`). Run `mpirun -np 48 lelphc -F lelphc.in`. **Same SAVE:** LetzElPhC and the BSE must read the same `SAVE` (one wavefunction gauge); redo both if the nscf is redone. The precision of LetzElPhC's build must match that of the `SAVE` (single or double).

4.4 yambo

- 2D: `CUTGeo= "slab z"` with the RIM. BSE: `BSEBands` is the band window (it fixes `--nv`), `BSEQptR` covers every irreducible $\mathbf{Q}$, `BSSmod= "d"` (full diagonalisation) and `WRbsWF` (eigenvectors written).
- Check that the energies in `ndb.BS_diago_Q1` are not all zero and that the report's `[K+QP]` gap equals the Kohn–Sham gap plus the GW correction.

5. Commands, in order

5.1 Archives

```
xphd ibz --savepath <SAVE>
xphd dmats --bse-dir <BSE_DIR> --savepath <SAVE> --ndb-elph <ELPH> --nexc <n>
xphd generate 0 <ARCH> --nexc <n> --savepath <SAVE> --ndb-elph <ELPH> --bse-dir <BSE_DIR> --dmats Dmats.npy
xphd check-archive <ARCH>/GI_ExcPh_Q0001.npz --detail
mpirun -np <R> xphd generate all <ARCH> --nexc <n> --savepath <SAVE> --ndb-elph <ELPH> \
       --bse-dir <BSE_DIR> --dmats Dmats.npy
xphd verify-archives --dir <ARCH> --n <N*N>
```

What `ibz` lists the irreducible $\mathbf{Q}$. `dmats` builds the single-particle rotation matrices over the BSE band window. `generate` computes $G(\mathbf{Q},\mathbf{q})$ for every $\mathbf{q}$, branch and pair of states, rotating excitons from the irreducible BSE; `0` is Γ, `all` the whole zone (shared between MPI ranks, resumable).

Needs `SAVE`, the BSE databases, `ndb.elph`.

Writes `Dmats.npy`; `GI_ExcPh_Q<iQ+1>.npz` per $\mathbf{Q}$.

Check `check-archive` prints READY, with time reversal and C_3 both near 10^{-3}; one far above the other is a rotation fault (or a split doublet: `--detail`, `--deg-tol`). `verify-archives`: all N^2 present and complete.

5.2 Phonons on the fine mesh

```
q2r.x < q2r.in
xphd matdyn-split --n <Nf> --chunks 48 --prefix qlist
for c in $(seq -f %03g 0 47); do                        # matdyn.x is serial: one process per chunk
  sed -e "s/FLVEC/modes_$c.modes/" -e "s/FLFRQ/freq_$c.freq/" matdyn.in.tmpl > in_$c
  cat qlist_$c.txt >> in_$c; matdyn.x < in_$c > out_$c &
done; wait
xphd matdyn-merge --chunks 48 -o matdyn.modes --freq matdyn.freq
xphd modes-check --modes matdyn.modes --freq matdyn.freq --n <Nf>
xphd matdyn matdyn.freq --fine <Nf> --nbnd <3 x atoms> --npz <ARCH>/GI_ExcPh_Q0001.npz -o hw_fine.npy
xphd mode-labels --modes matdyn.modes --fine <Nf> --mesh <N> --masses <m_1> <m_2> \
                 --archive <ARCH>/GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy -o mode_labels.npy
```

What The fine-mesh frequencies in eV, row-major; `mode-labels` gives each branch its in-/out-of-plane character on the archive mesh (planar two-atom layers; for layers where σ_h exchanges atoms take the parities from the el–ph database).

Needs The force constants; one archive for its $\mathbf{q}$ ordering.

Writes `matdyn.modes`, `matdyn.freq`, `hw_fine.npy`, `mode_labels.npy`.

Check `modes-check` passes; `matdyn` reproduces the archive's frequencies where the meshes coincide (μeV). Negative frequencies only next to Γ in the flexural branch.

5.3 Linewidths, self-energy and lineshape

```
xphd linewidth <ARCH>/GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --T 10 77 300 \
      --acoustic-model q2 --n-shell 1 --per-mode -o G.npz
xphd sweep '<ARCH>/GI_ExcPh_Q*.npz' --hw-fine hw_fine.npy --T 77 --acoustic-model q2 --n-shell 1 \
      --workers 8 --unfold <SAVE> -o lw_77K.npz
xphd selfenergy <ARCH>/GI_ExcPh_Q0001.npz --hw-fine hw_fine.npy --state 1 --T 300 -o se.npz
xphd lineshape <ARCH>/GI_ExcPh_Q0001.npz --T 10 77 300 --hw-fine hw_fine.npy \
      --acoustic-model shell --acoustic-scaling omega --parity parity.npz \
      --bsdb <BSE_DIR>/ndb.BS_diago_Q1 --n-omega 4801 -o lineshape.npz
```

What `linewidth`: golden-rule width of every state at one $\mathbf{Q}$, the energy δ integrated on fine-mesh triangles (exciton energies refined by local quadratics), emission/absorption and per branch. `sweep`: the same at every $\mathbf{Q}$, unfolded onto the zone. `selfenergy`: $\Sigma(\omega)$ of one state with its Kramers–Kronig partner. `lineshape`: the full Σ matrix and the absorption line; final states carry the self-consistent width (`--final-width sc`, default).

Needs Archives and `hw_fine.npy`; for `lineshape` the dipoles (`--bsdb` or `--dipoles`) and `parity.npz`.

Writes `G.npz`, `lw_77K.npz`, `se.npz`, `lineshape.npz`.

Check The acoustic channel inside the cell around $q = 0$ is zero by the sum rule unless a model is chosen: `none` drops it, which underestimates the width at a band minimum. `lineshape` prints the symmetry check (opposite parities ≈ 0, degenerate blocks proportional to the identity), the final-width gain (well below 1) and warns if the line is narrower than half its on-shell width; the widths at low T need a frequency step below the width (`--n-omega`).

5.4 Symmetry

```
xphd parity --save <SAVE> --bse-dir <BSE_DIR> --dmats Dmats.npy --nv <nv> --nstates <n> \
            --deg-tol 0.005 -o parity.npz
xphd band-parity --save <SAVE> --bse-dir <BSE_DIR> --dmats Dmats.npy --nv <nv> --path G M K G -o band_parity.npz
xphd irreps --save <SAVE> --bse-dir <BSE_DIR> --dmats Dmats.npy --nv <nv> --points G K M \
            --nstates <n> -o irreps.npz
xphd selection-rule --archive <ARCH>/GI_ExcPh_Q0001.npz --parity parity.npz --labels mode_labels.npy \
                    --state 1 --by-final
```

What $\chi(\sigma_h)$ of every exciton at every $\mathbf{Q}$ (unfolded with yambo's own map); band parities; irrep labels with the closure of each multiplet; the coupling split into symmetry-allowed and -forbidden.

Needs `SAVE`, the BSE at all $\mathbf{Q}$, `Dmats.npy`, an archive, phonon labels.

Writes `parity.npz`, `band_parity.npz`, `irreps.npz`, figures.

Check Every state ± 1 (fractional values: a mixed near-degenerate pair; `--rotate` in `selection-rule`); forbidden/allowed at the level of numerical noise; every multiplet closes.

5.5 Optics

```
xphd dipoles --save <SAVE> --bse-dir <BSE_DIR> -o exc_dipoles.npy --deg-tol 0.005
xphd chirality --save <SAVE> --bse-dir <BSE_DIR> --dipoles exc_dipoles.npy --states 1 2
```

What Complex x, y, z dipoles of the $\mathbf{Q} = 0$ states from one BSE (validated against yambo's residuals); helicity and valley polarisation of a degenerate pair.

Needs `SAVE`, the BSE with `ndb.dipoles`.

Writes `exc_dipoles.npy`; numbers.

Check Residuals agree per degenerate multiplet; a bright valley pair has helicity ± 1.

5.6 Transport

```
xphd bte '<ARCH>/GI_ExcPh_Q*.npz' --T 77 --hw-fine hw_fine.npy --parity parity.npz --window 0.1 --plan
xphd bte '<ARCH>/GI_ExcPh_Q*.npz' --T 77 --hw-fine hw_fine.npy --parity parity.npz --window 0.1 \
     --acoustic-model shell --acoustic-scaling omega --inject uniform --state 3 4 \
     --t-end 20000 --nt 200 --workers 8 --snapshots snaps_77K.npz
```

What Populations of every state at every $\mathbf{Q}$ under the full rate matrix (detailed balance, out-rates refined on the fine mesh). `--window` keeps the states within that energy of the minimum on the refined mesh (`--plan` prints them and the memory first). `--inject`: `optical` (oscillator-weighted, optional `--pump-center`/`--pump-sigma`), `thermal`, or `uniform` over `--state` (a degenerate bright pair: the basis-independent unpolarised pump).
Needs Every archive of the zone, `hw_fine.npy`; `parity.npz` with `--window`.
Writes The snapshot file (populations, energies, per-state widths at every delay).
Check $N/N_0 = 1$ throughout; the final distribution against Boltzmann at T (printed).

5.7 Photoemission

```
xphd arpes --save <SAVE> --bse-dir <BSE_DIR> --qp-db <GW>/ndb.QP --temperature 300 --nstates 100 -o arpes_300K.npz
xphd arpes --save <SAVE> --bse-dir <BSE_DIR> --qp-db <GW>/ndb.QP --snapshots snaps_77K.npz \
           --times 0 1000 10000 --path M K G "K'" M --sigma 0.01 --box K 0.12 --valence \
           -o arpes/arpes_{t}fs.npz --series arpes/series.npz
```

What $I(\mathbf{k}, E) = \sum N_{S\mathbf{Q}} \sum_{vc} |A^{S\mathbf{Q}}_{vc\mathbf{k}}|^2 \, \delta(E - E_{S\mathbf{Q}} - \varepsilon_{v,\mathbf{k}-\mathbf{Q}})$ with Boltzmann (`--temperature`) or transport (`--snapshots`) occupations; maps on a path, momentum boxes, the valence band with its holes, the spectra against the delay.
Needs `SAVE`, the BSE, `ndb.QP`; a snapshot file.
Writes One map per delay and the series file.
Check The replica lies below the conduction band by the binding energy; the zone-integrated intensity is constant in time. With spin–orbit bands use `--spin-deg 1`. In PowerShell write `"K'"` in double quotes.

6. The run report

After every command, also after a failure, rank 0 writes a report: `<output>_report.pdf` next to the main output, or, when the run wrote nothing, `xphd_<command>_<time>_report.pdf` in the working directory (`.txt` on Linux, below). The PDF holds:

- a title bar naming the step (BTE, ARPES, Linewidths, . . .) with the version;
- the xphd banner printed at the start of the run (version, developers and their affiliations, how to cite, acknowledgements), beside a diagram of the pipeline (Quantum ESPRESSO, LetzElPhC, yambo and xphd) with this command's step marked;
- the command line, status, start time and wall time, working directory;
- the environment: xphd, Python and library versions, host, MPI ranks, thread settings;
- every input file named on the command line with size, date and a SHA-256 checksum (files up to 64 MB);
- every parameter, defaults included;
- every file written or changed by the run;
- plots of the main results, for the outputs of `generate`, `parity`, `linewidth`, `sweep`, `lineshape`, `bte` and `arpes`, each figure with lettered, titled panels and a caption saying what is drawn; quantities over the zone are shown along Γ–M–K–Γ (the mesh points on the path) and over the hexagonal first Brillouin zone. Only the essential panels are drawn. An archive: the phonon dispersion, the final exciton bands with the one-phonon window, the coupling out of the lowest state over the zone and per branch along the path. `parity`: the exciton bands coloured by σ_h parity. `linewidth`: the width of every state, and of state 1 by branch. `sweep`: the width of the lowest exciton over the zone and along the path. `lineshape`: the width of the A line and ε_2. `bte`: the mean energy, the energy distribution and the population over the zone. `arpes`: the map along the path and the occupied excitons against their momentum; against the delay, the whole-zone spectrum and the mean exciton energy. Other files, such as `Dmats.npy`, are listed and not drawn;
- the full log as the terminal leaves it (a progress bar is one line, its final state), and the error if the run failed.

Off with `xphd --no-report <command> ...` or `XPHD_REPORT=0`.

On Linux and clusters. The system decides the kind of report: the PDF on Windows and macOS, a text report, `<output>_report.txt`, on Linux (a cluster or an Ubuntu machine), where a terminal has no PDF viewer. The text report holds the same banner, run record, inputs with checksums, parameters, outputs and full log, everything but the plots:

```
less lw_report.txt                                  # on the cluster
```

`XPHD_REPORT=pdf`, `txt` or `both` overrides the choice. With `both` on a cluster the PDF is drawn too (no display is needed; DejaVu fonts where the desktop ones are missing) and the text report lists its figures by title, panels and caption; copy it home for the plots (`scp user@cluster:/path/to/run/lw_report.pdf .`, or WinSCP). The PDF needs `matplotlib` (the `plot` extra, installed by `scripts/hpc/install_cluster.sh`); without it the text report is written.

7. Running in parallel

The numbers do not depend on the layout: serial, process-pool and MPI runs write identical arrays.

Command	Spread over	Memory per process
`generate`	MPI ranks over $\mathbf{Q}$; BLAS threads in each rank	one copy of the el–ph and BSE databases
`sweep`	MPI ranks (one per core) or `--workers`	one archive and the fine mesh
`bte`, out-rate refinement	MPI ranks or `--workers`	bounded by `--mem-per-task` / `XPHD_TASK_MEM_GB`
`bte`, propagation	one process, every core as BLAS threads	about $10 \times (\text{states})^2 \times 8$ bytes
every other command	rank 0 only	

Install on a cluster (`scripts/hpc/install_cluster.sh` does this): `mpi4py` must be built against the MPI that launches the job; the PyPI wheel brings its own MPI and runs unconnected copies.

```
module load <compilers and MPI used for QE/yambo>
python3 -m venv ~/xphd-env && source ~/xphd-env/bin/activate
MPICC=$(which mpicc) pip install --no-binary=mpi4py mpi4py      # Intel MPI: MPICC=mpiicc
pip install -e "./xphd[hpc,yambo,plot]"
mpirun -np 4 xphd hpc-info                                      # ranks, nodes, cores, BLAS threads, memory
```

Sizing. `generate`: measure one $\mathbf{Q}$ (`/usr/bin/time -v`, maximum resident size); ranks per node $= \lfloor 0.85\times$ node memory / that size$\rfloor$, cores per rank as BLAS threads. `sweep` and the `bte` refinement: one rank per core, `OMP_NUM_THREADS=1`; more ranks than $\mathbf{Q}$ points gain nothing. `bte` propagation: one node, all cores as threads (`--threads`); unbind it (`srun --cpu-bind=none`, `mpirun --bind-to none`). `generate --resume` (default for several $\mathbf{Q}$) and `--checkpoint DIR` (`sweep`, `bte`) make a stopped job resumable: resubmit the same script. SLURM and PBS templates for every stage are in `scripts/hpc/`.

Variable	Meaning
`OMP_NUM_THREADS`	BLAS threads per process (`--threads`, `--blas-threads` override)
`XPHD_TASK_MEM_GB`	memory bound per `bte` refinement task (default 2)
`XPHD_MPI=0 / 1`	disable / force MPI detection
`XPHD_ALL_RANKS=1`	every rank prints (debugging)
`XPHD_MP_START`	start method for `--workers`
`XPHD_NO_BANNER=1`	no start-up banner
`XPHD_REPORT=0`	no report
`XPHD_REPORT=pdf / txt / both`	which report; by default the PDF on Windows and macOS, the text report on Linux

8. When a result looks wrong

Symptom	Cause	Fix
Every BSE energy exactly 0, eigenvectors fine	a compiler fault in yambo's `BS_E=cmplx(BS_E_real,...)` (Intel ifort)	the element loop in `K_diago_driver.F`
LetzElPhC: "Unable to open the dyn0 file"	`ph_save_dir` not set	`ph_save_dir = ph.save`
dvscf files of 0 bytes after a `ph.x` restart	`recover` trusted old `phsave` records	delete those `dynmat.<iq>.*.xml`, rerun the $\mathbf{q}$ without `recover`
`check-archive`: one of TR/C_3 far above the other	a rotation fault, or a doublet split by long-range exchange	`--detail`; raise `--deg-tol`
Fractional parities in a ring of $\mathbf{Q}$	mixed near-degenerate states of opposite parity	`parity --deg-tol`, `selection-rule --rotate`
Width at a band minimum far too small	the acoustic channel inside the first cell is missing	`--acoustic-model q2` (`linewidth`) or `shell` (`lineshape`, `bte`)
Band widths too steep near crossings	global Fourier interpolation rings	`--band-interp local` (default)
`matdyn` outputs broken after `cat`	a header and cumulative distances per chunk	`xphd matdyn-merge`
ε_2 line much narrower than its width at E_A	one-shot Σ at the band bottom	`lineshape --final-width sc` (default)
yambopy "open-shell" error reading `ndb.dipoles`	band count when the first band is not 1	`xphd dipoles`
`hpc-info`: "MPI not active" under `srun`	`mpi4py` not built against this MPI, or `srun` without `--mpi=pmix`	rebuild `mpi4py` from source; `srun --mpi=pmix`

Symptom	Cause	Fix
“launched with N MPI ranks but mpi4py cannot be imported”	the environment is not activated in the job script	`source ~/xphd-env/bin/activate` in the script
K/K′ weights differ transiently in `bte`	the solver’s basis inside a degenerate pair	`--inject uniform --state <pair>`, `--deg-tol`

A. Command reference

Generated from `xphd <command> --help`.

Archives

xphd ibz

```
usage: xphd ibz [-h] [--savepath SAVEPATH] [--mesh MESH MESH] [--iq-map IQ_MAP]

options:
  -h, --help           show this help message and exit
  --savepath SAVEPATH  directory holding ns.db1
  --mesh MESH MESH     only for the fallback when yambopy is unavailable
  --iq-map IQ_MAP      fallback only: an archive whose q_red ordering to report the points in
```

xphd dmats

```
usage: xphd dmats [-h] [--bse-dir BSE_DIR] [--savepath SAVEPATH] [--ndb-elph NDB_ELPH]
                  [--nexc NEXC] [--tag TAG]

options:
  -h, --help           show this help message and exit
  --bse-dir BSE_DIR
  --savepath SAVEPATH
  --ndb-elph NDB_ELPH
  --nexc NEXC
  --tag TAG            prefix for the reference file
```

xphd generate

```
usage: xphd generate [-h] [--threads THREADS] [--verify-threads] [--blas-threads BLAS_THREADS]
                     [--resume | --overwrite] [--mesh MESH MESH] [--nexc NEXC]
                     [--savepath SAVEPATH] [--ndb-elph NDB_ELPH] [--bse-dir BSE_DIR]
                     [--dmats DMATS] [--allow-no-split]
                     iQ [outdir]

Exciton-phonon archives from LetzElPhC + yambo. One Q, a list, or the whole zone; under
mpirun/srun the list is shared between the ranks and each rank loads the databases ONCE for all of
its points.

positional arguments:
  iQ                    0-based index in the full BZ: '17', a list '0,4,9-12', or 'all'
  outdir

options:
  -h, --help            show this help message and exit
  --threads THREADS     1 = serial (safe). >1 shares a netCDF handle across threads; run --verify-
                        threads first.
  --verify-threads      run 8 q-points serially and threaded, assert identical
  --blas-threads BLAS_THREADS
                        BLAS/OpenMP threads for the contractions inside yambopy (default:
                        OMP_NUM_THREADS as set)
  --resume              skip archives that already exist and are complete (default when more than
                        one Q is requested)
  --overwrite           regenerate even complete archives
  --mesh MESH MESH      n1 n2 of the k/q mesh; derived from the lattice when omitted
  --nexc NEXC           exciton states per Q to keep
  --savepath SAVEPATH
  --ndb-elph NDB_ELPH
  --bse-dir BSE_DIR
  --dmats DMATS
  --allow-no-split      accept an archive without Ge_grid/Gh_grid if the electron/hole flags
                        cannot be found. Off by default: the interference analysis needs them, and
                        a silent omission is only discovered after every Q has run.
```

xphd check-archive

```
usage: xphd check-archive [-h] [--detail] [--deg-tol DEG_TOL] archive

Validate ONE archive before generating the rest.

positional arguments:
  archive
```

```
options:
  -h, --help         show this help message and exit
  --detail           attribute any symmetry mismatch to phonon branches, initial and final states,
                     and list the worst q-points
  --deg-tol DEG_TOL  eV; states closer than this form one multiplet in the symmetry test (default
                     15 meV). At Q = 0 the BSE's long-range exchange, taken along one direction,
                     splits each E' doublet into a longitudinal and a transverse member -- by 10
                     meV for GaN's second bright pair -- and each member alone is not
                     C3-symmetric; only their sum is. Grouping more states than strictly
                     degenerate only sums symmetric quantities, so a generous value is safe
```

xphd verify-archives

```
usage: xphd verify-archives [-h] [--dir DIR] [--n N]

options:
  -h, --help  show this help message and exit
  --dir DIR
  --n N
```

xphd check

```
usage: xphd check [-h] [--mesh MESH MESH] npz

positional arguments:
  npz

options:
  -h, --help        show this help message and exit
  --mesh MESH MESH
```

Phonons on the fine mesh

xphd matdyn-split

```
usage: xphd matdyn-split [-h] [--n N] [--chunks CHUNKS] [--prefix PREFIX]

options:
  -h, --help       show this help message and exit
  --n N            fine mesh is n x n x 1
  --chunks CHUNKS  number of pieces; match the cores on one node, or the array size you intend to
                   submit
  --prefix PREFIX
```

xphd matdyn-merge

```
usage: xphd matdyn-merge [-h] [--chunks CHUNKS] [--modes-prefix MODES_PREFIX]
                         [--freq-prefix FREQ_PREFIX] [-o OUT_MODES] [--freq FREQ] [--index INDEX]

options:
  -h, --help            show this help message and exit
  --chunks CHUNKS
  --modes-prefix MODES_PREFIX
  --freq-prefix FREQ_PREFIX
  -o, --out-modes OUT_MODES
  --freq FREQ
  --index INDEX         written by xphd matdyn-split; used to check that the merged length is what
                        was requested
```

xphd modes-check

```
usage: xphd modes-check [-h] [--modes MODES] [--freq FREQ] [--n N]

options:
  -h, --help     show this help message and exit
  --modes MODES
  --freq FREQ
  --n N          expected mesh
```

xphd matdyn

```
usage: xphd matdyn [-h] [--fine FINE] [--nbnd NBND] [--npz NPZ] [--mesh MESH MESH]
                   [--q-order {rowmajor,npz}] [-o OUT]
                   freq

positional arguments:
  freq

options:
  -h, --help            show this help message and exit
  --fine FINE
  --nbnd NBND
```

```
  --npz NPZ
  --mesh MESH MESH
  --q-order {rowmajor,npz}
  -o, --out OUT
```

xphd mode-labels

```
usage: xphd mode-labels [-h] [--modes MODES] [--fine FINE] [--mesh MESH]
                        --masses MASSES [MASSES ...] [--archive ARCHIVE] [--hw-fine HW_FINE]
                        [-o OUT]

Phonon branch characters on the archive mesh -> mode_labels.npy.

options:
  -h, --help            show this help message and exit
  --modes MODES         matdyn eigenvectors on the refined mesh
  --fine FINE           the refined mesh is fine x fine
  --mesh MESH           the source (archive) mesh is mesh x mesh
  --masses MASSES [MASSES ...]
                        atomic masses in amu, in the order of the cell. No default: GaN 69.723
                        14.007, hBN 10.81 14.007
  --archive ARCHIVE     any archive; only its q_red ordering is used
  --hw-fine HW_FINE     refined frequencies from xphd matdyn, to cross-check against the
                        frequencies in --modes
  -o, --out OUT
```

Linewidths, self-energy and lineshape

xphd linewidth

```
usage: xphd linewidth [-h] [--mesh MESH MESH] [--hw-fine HW_FINE] [--fine FINE] [--refine REFINE]
                      [--scheme {const,fft}] [--T T [T ...]] [--flat-width FLAT_WIDTH]
                      [--acoustic-cut ACOUSTIC_CUT] [--acoustic-model {none,q2,auto}]
                      [--n-shell N_SHELL] [--band-interp {local,fourier}] [-o OUT] [--per-mode]
                      npz

positional arguments:
  npz

options:
  -h, --help            show this help message and exit
  --mesh MESH MESH
  --hw-fine HW_FINE
  --fine FINE
  --refine REFINE
  --scheme {const,fft}
  --T T [T ...]
  --flat-width FLAT_WIDTH
  --acoustic-cut ACOUSTIC_CUT
  --acoustic-model {none,q2,auto}
                        acoustic coupling inside the cell around q = 0, where the constant-
                        coupling value is zero by the sum rule. none: leave it zero, which drops
                        small-q acoustic scattering -- the ONLY channel at a band minimum. q2:
                        |G|^2 * 2w = C q^2, the form for a neutral exciton; use this. auto: also
                        keep a constant A where it is above 2 sigma -- right for free carriers,
                        but for an exciton a kept A is a fitting artefact and is flagged
  --n-shell N_SHELL     innermost shells of the source mesh the fit uses; at least 1 for q2, 3 for
                        auto. The per-shell C it prints shows whether the result depends on the
                        choice
  --band-interp {local,fourier}
                        how the exciton energies are refined onto the fine mesh. local (default):
                        a quadratic around every mesh point from its six neighbours, which cannot
                        ring. fourier: the global interpolant, which rings from every band
                        crossing -- at GaN's K minimum it made the band 34% too steep. Reproduces
                        earlier results
  -o, --out OUT
  --per-mode            also print linewidths by phonon branch; with a text -o, write
                        <stem>_modes.txt as well
```

xphd sweep

```
usage: xphd sweep [-h] [--mesh MESH MESH] [--hw-fine HW_FINE] [--fine FINE] [--refine REFINE]
                  [--scheme {const,fft}] [--T T [T ...]] [--flat-width FLAT_WIDTH]
                  [--acoustic-cut ACOUSTIC_CUT] [--workers WORKERS] [--checkpoint DIR]
                  [--unfold SAVE] [--legacy] [-o OUT] [--acoustic-model {none,q2,auto}]
                  [--n-shell N_SHELL] [--band-interp {local,fourier}]
                  pattern

positional arguments:
  pattern               glob, e.g. 'GI_ExcPh_Q*.npz'

options:
  -h, --help            show this help message and exit
  --mesh MESH MESH
  --hw-fine HW_FINE
  --fine FINE
  --refine REFINE
  --scheme {const,fft}
```

```
  --T T [T ...]
  --flat-width FLAT_WIDTH
  --acoustic-cut ACOUSTIC_CUT
  --workers WORKERS     local processes (per MPI rank under mpirun/srun)
  --checkpoint DIR      save each finished Q in DIR; a rerun skips them
  --unfold SAVE
  --legacy              also write Qx, Qy and a 2D LW (nexc, nq) in meV, as aggregate.py did, for
                        existing plotting scripts
  -o, --out OUT
  --acoustic-model {none,q2,auto}
                        small-q acoustic coupling, as in xphd linewidth; use q2, and the same
                        --n-shell as the single-Q runs
  --n-shell N_SHELL
  --band-interp {local,fourier}
                        how the exciton energies are refined onto the fine mesh. local (default):
                        a quadratic around every mesh point from its six neighbours, which cannot
                        ring. fourier: the global interpolant, which rings from every band
                        crossing -- at GaN's K minimum it made the band 34% too steep. Reproduces
                        earlier results
```

xphd unfold

```
usage: xphd unfold [-h] [--savepath SAVEPATH] [--mesh MESH MESH] [--legacy] [-o OUT] field

positional arguments:
  field

options:
  -h, --help           show this help message and exit
  --savepath SAVEPATH
  --mesh MESH MESH
  --legacy             also write Qx, Qy and a 2D LW (nexc, nq) in meV
  -o, --out OUT
```

xphd selfenergy

```
usage: xphd selfenergy [-h] [--mesh MESH MESH] [--state STATE] [--T T] [--hw-fine HW_FINE]
                       [--refine REFINE] [--window WINDOW] [--n-omega N_OMEGA]
                       [--acoustic-cut ACOUSTIC_CUT] [-o OUT] [--band-interp {local,fourier}]
                       npz

positional arguments:
  npz

options:
  -h, --help            show this help message and exit
  --mesh MESH MESH
  --state STATE
  --T T
  --hw-fine HW_FINE
  --refine REFINE       the Kramers-Kronig transform integrates over omega and does not need the
                        on-shell resolution; a coarser refinement here is usually enough
  --window WINDOW       eV, half-width about the bare energy. 0 (default) spans the full support
                        of Im Sigma instead, which is what Kramers-Kronig needs: Im Sigma is one-
                        sided for states near the bottom of the band, so a symmetric window cuts
                        through its maximum.
  --n-omega N_OMEGA
  --acoustic-cut ACOUSTIC_CUT
  -o, --out OUT
  --band-interp {local,fourier}
                        exciton-energy refinement, as in xphd linewidth
```

xphd lineshape

```
usage: xphd lineshape [-h] [--mesh MESH MESH] [--T T [T ...]] [--hw-fine HW_FINE]
                      [--refine REFINE] [--n-omega N_OMEGA] [--pad PAD]
                      [--acoustic-cut ACOUSTIC_CUT] [--acoustic-model {none,shell}]
                      [--acoustic-scaling {none,omega}] [--parity PARITY] [--bsdb BSDB]
                      [--cells {hex,rhombic}] [--dipoles DIPOLES] [--eta ETA]
                      [--window WINDOW WINDOW] [--n-plot N_PLOT] [--final-width FINAL_WIDTH]
                      [--no-recenter] [-o OUT] [--band-interp {local,fourier}]
                      npz

positional arguments:
  npz

options:
  -h, --help            show this help message and exit
  --mesh MESH MESH
  --T T [T ...]         K; several temperatures share one pass over the grid
  --hw-fine HW_FINE
  --refine REFINE
  --n-omega N_OMEGA     points of the uniform self-energy grid (Kramers-Kronig); keep its step <=
                        ~1 meV
  --pad PAD             eV beyond the support of Im Sigma. The grid already spans the final-state
                        range plus the largest phonon; this is extra margin.
  --acoustic-cut ACOUSTIC_CUT
```

```
  --acoustic-model {none,shell}
                        the q = 0 cell for the acoustic branches: 'shell' fills it from the first
                        shell between the same states (needs --parity), as xphd bte --window does
  --acoustic-scaling {none,omega}
                        'omega' carries acoustic couplings to the fine q as g^2 ~ w, as xphd bte
                        --acoustic-scaling omega
  --parity PARITY       exciton_parity.npz (sigma_h parity of every state at every Q), for
                        --acoustic-model shell and the symmetry report
  --bsdb BSDB           ndb.BS_diago_Q1 of the SAME diagonalisation as the archive, for the
                        dipoles and hence eps_2
  --cells {hex,rhombic}
                        which coarse point a fine point takes its coupling and energy fit from:
                        its Wigner-Seitz hexagon (default; keeps the point-group symmetry of
                        Sigma) or the rounded reduced coordinates (the BTE's and xphd linewidth's
                        cells, not C3-invariant)
  --dipoles DIPOLES     the dipoles as a .npy (complex, one per state) instead of --bsdb, e.g.
                        saved with lineshape.read_dipoles
  --eta ETA             eV, added to every width; keep it far below the narrowest
  --window WINDOW WINDOW
                        eV, where to look for the A peak (default: from the lowest bright
                        multiplet to half way to the next)
  --n-plot N_PLOT       points of the saved eps_2 curves
  --final-width FINAL_WIDTH
                        half-width given to every final state, Sigma(w) -> Sigma(w + i eta): 'sc'
                        (default) equal to half the A width it produces (self-consistent), or a
                        value in eV; 0 is one-shot. The A line sits at the bottom of its own band,
                        where the acoustic channel makes Im Sigma step up within a fraction of a
                        meV: one-shot, the on-shell width depends on where E_A falls on the
                        frequency grid and the eps_2 line locks below the step
  --no-recenter         compute eps_2 with the A line where Re Sigma puts it; by default the
                        static A shift is taken off the whole matrix (all excitons renormalised
                        together)
  -o, --out OUT
  --band-interp {local,fourier}
                        exciton-energy refinement, as in xphd linewidth
```

Symmetry

xphd parity

```
usage: xphd parity [-h] [--save SAVE] [--bse-dir BSE_DIR] [--dmats DMATS] --nv NV
                   [--nstates NSTATES] [--deg-tol DEG_TOL] [-o OUT]

chi(sigma_h) for every exciton state at every Q.

options:
  -h, --help         show this help message and exit
  --save SAVE
  --bse-dir BSE_DIR
  --dmats DMATS
  --nv NV            valence bands in the BSE window (3 for hBN and GaN)
  --nstates NSTATES  states per Q; 1 is enough for the lowest-band map and is far faster
  --deg-tol DEG_TOL  eV; states closer than this form one multiplet, whose sigma_h matrix is
                     stored so that `xphd selection-rule --rotate` can split the coupling by the
                     multiplet's parity eigenstates instead of the solver's mixed states (default
                     5 meV; 0 stores none)
  -o, --out OUT
```

xphd band-parity

```
usage: xphd band-parity [-h] [--save SAVE] [--bse-dir BSE_DIR] [--dmats DMATS] --nv NV
                        [--path PATH [PATH ...]] [--tol TOL] [-o OUT]

Mirror parity of the electronic bands, along a path and over the zone.

options:
  -h, --help            show this help message and exit
  --save SAVE           directory holding ns.db1
  --bse-dir BSE_DIR     for the band window, read from ndb.BS_diago_Q1
  --dmats DMATS
  --nv NV               valence bands inside the BSE window
  --path PATH [PATH ...]
                        high-symmetry labels, as in xphd irreps
  --tol TOL             how far from +-1 a parity may sit and still count
  -o, --out OUT
```

xphd irreps

```
usage: xphd irreps [-h] [--save SAVE] [--bse-dir BSE_DIR] [--dmats DMATS] --nv NV
                   [--points [POINTS ...]] [--nstates NSTATES] [--deg-tol DEG_TOL]
                   [--closure-tol CLOSURE_TOL] [-o OUT]

Irreducible-representation labels at the high-symmetry points.

options:
  -h, --help            show this help message and exit
  --save SAVE
  --bse-dir BSE_DIR
```

```
  --dmats DMATS
  --nv NV
  --points [POINTS ...]
                        any of: G, K, K', M, M', M'', Q, Q'
  --nstates NSTATES
  --deg-tol DEG_TOL     eV; states closer than this form one multiplet
  --closure-tol CLOSURE_TOL
                        largest |D(R)D(S) - D(RS)| accepted for a multiplet (default 1e-2)
  -o, --out OUT         write the table to an .npz (per point: energies, dimensions, chi(sh),
                        labels, residuals, closure, characters), e.g. for
                        scripts/compare_mos2_refs.py
```

xphd selection-rule

```
usage: xphd selection-rule [-h] [--archive ARCHIVE] [--parity PARITY] [--labels LABELS]
                           [--state STATE] [--tol TOL] [--rotate] [--by-final] [--nfinal NFINAL]
                           [--modes-for MODES_FOR] [--clip CLIP] [--dpi DPI] [--out OUT]

options:
  -h, --help            show this help message and exit
  --archive ARCHIVE
  --parity PARITY
  --labels LABELS
  --state STATE         initial exciton
  --tol TOL             |chi| below this is treated as undefined and the point is excluded rather
                        than forced to a parity
  --rotate              split the coupling by the parity EIGENSTATES of each near-degenerate
                        multiplet of final states (needs a parity.npz written with --deg-tol),
                        instead of by the per-state chi of the solver's possibly mixed states.
                        Nothing is excluded for being mixed; only eigenvectors with |mu| < --tol
                        remain undefined
  --by-final            also map the coupling resolved by final state
  --nfinal NFINAL
  --modes-for MODES_FOR
                        final state index; adds one panel per phonon branch for that channel
  --clip CLIP
  --dpi DPI
  --out OUT
```

Optics

xphd dipoles

```
usage: xphd dipoles [-h] [--save SAVE] [--bse-dir BSE_DIR] [--dipoles-dir DIPOLES_DIR] [-o OUT]
                    [--window WINDOW] [--deg-tol DEG_TOL] [--compare COMPARE]

Exciton dipoles for emission, written where exc_ph_get_inputs finds them.

options:
  -h, --help            show this help message and exit
  --save SAVE
  --bse-dir BSE_DIR
  --dipoles-dir DIPOLES_DIR
                        directory holding ndb.dipoles; defaults to --bse-dir
  -o, --out OUT
  --window WINDOW       eV above the lowest exciton for the residual check
  --deg-tol DEG_TOL
  --compare COMPARE     a reference exc_dipoles.npy, e.g. GaN's from the working yambopy path, to
                        compare |D|^2 against
```

xphd chirality

```
usage: xphd chirality [-h] [--save SAVE] [--bse-dir BSE_DIR] [--dipoles DIPOLES]
                      --states STATES STATES [--out OUT]

Chirality of a degenerate exciton pair: helicity and valley polarisation.

options:
  -h, --help            show this help message and exit
  --save SAVE
  --bse-dir BSE_DIR
  --dipoles DIPOLES     written by `xphd dipoles`
  --states STATES STATES
                        1-based indices of the degenerate pair
  --out OUT
```

Transport and photoemission

xphd bte

```
usage: xphd bte [-h] [--mesh MESH MESH] [--T T] [--acoustic-cut ACOUSTIC_CUT]
                [--flat-width FLAT_WIDTH] [--refine REFINE] [--window EV]
                [--band-interp {local,fourier}] [--acoustic-model {none,q2,shell}]
                [--parity PARITY] [--deg-tol DEG_TOL] [--n-shell N_SHELL]
                [--acoustic-scaling {none,omega,power}] [--acoustic-power ACOUSTIC_POWER] [--plan]
                [--hw-fine HW_FINE] [--scheme {const,complexfft}] [--bose {node,crossing}]
```

```
               [--inject {optical,thermal,uniform}] [--state [STATE ...]]
               [--pump-center PUMP_CENTER] [--pump-sigma PUMP_SIGMA] [--hot HOT] [--width WIDTH]
               [--density DENSITY] [--t-end T_END] [--nt NT] [--workers WORKERS]
               [--threads THREADS] [--checkpoint DIR] [--mem-per-task GB] [--refine-only]
               [--save-outrates FILE] [--outrates FILE] [--jac {analytic,fd}]
               [--matrix {dense,sparse,auto}] [--rtol RTOL] [--atol ATOL] [--channels]
               [--snapshots SNAPSHOTS]
               pattern

positional arguments:
  pattern               glob, e.g. 'GI_ExcPh_Q*.npz'

options:
  -h, --help            show this help message and exit
  --mesh MESH MESH
  --T T                 lattice temperature
  --acoustic-cut ACOUSTIC_CUT
  --flat-width FLAT_WIDTH
                        eV; default 0 for the mesh matrix, 2.4e-3/(n f) with --window (as the
                        refined out-rates)
  --refine REFINE       refine the OUT-RATE integral and rescale rows. The matrix itself cannot be
                        refined (each q is a state, so its size grows as f^4) but the row sum can
                        -- or, with --window, the matrix on the refined mesh for the bottom of the
                        valleys only.
  --window EV           build the matrix on the REFINED mesh (--hw-fine / --refine) for the states
                        within EV of the lowest exciton, instead of the mesh matrix rescaled to
                        the refined out-rates. For valleys too steep for the source mesh. The
                        Boltzmann weight left outside is printed; --hw-fine may be finer than
                        --refine (subsampled exactly)
  --band-interp {local,fourier}
                        exciton energies on the refined mesh, with --window (as in xphd linewidth)
  --acoustic-model {none,q2,shell}
                        with --window: acoustic coupling between fine states with the same parent
                        (the cell around q = 0). none: zero (as the refined out-rates); q2: the
                        fitted C q^2 form of xphd linewidth (by band index: unsafe where bands of
                        opposite parity reorder); shell: the first-shell coupling between the same
                        states, matched by parity (needs --parity), carried down by --acoustic-
                        scaling
  --parity PARITY       parity.npz of the exciton states (Q_red, chi, E): needed by --window
                        --acoustic-model shell; with it, degenerate multiplets (--deg-tol) never
                        join opposite parities
  --deg-tol DEG_TOL     eV: states of one Q closer than this form a degenerate multiplet, over
                        which the couplings are averaged, so the populations do not depend on the
                        basis the BSE solver picked inside it (0: off)
  --n-shell N_SHELL     shells for --acoustic-model q2
  --acoustic-scaling {none,omega,power}
                        with --window: how an acoustic branch's coarse coupling is carried to the
                        fine q. none: constant; omega: g^2 ~ w, the long-wavelength law for strain
                        and curvature coupling; power: |M|^2 ~ q^n (--acoustic-power n) with the
                        actual fine w. For a soft quadratic branch (ZA) 'none' makes the rate
                        follow --acoustic-cut; see xphd.bte.windowed_rates
  --acoustic-power ACOUSTIC_POWER
                        n of |M|^2 ~ q^n for --acoustic-scaling power (ZA: 2 tilt, 4 curvature;
                        za_coupling_law.py measures it)
  --plan                with --window: print the states kept, the memory, the Boltzmann weight
                        left out and the energy step against kT, then stop (seconds; use it to
                        pick --refine and --window)
  --hw-fine HW_FINE     matdyn frequencies on the refined mesh, as used by the linewidths. Sets
                        the refinement and makes the two calculations use identical phonons.
  --scheme {const,complexfft}
  --bose {node,crossing}
                        phonon occupation in the rate matrix: node = at the energy difference of
                        the two mesh states; crossing = each phonon's own occupation where the
                        delta crossing happens, as the refined out-rates use. Default: node for
                        the mesh matrix (as before), crossing with --window (see
                        xphd.bte.windowed_rates)
  --inject {optical,thermal,uniform}
  --state [STATE ...]   1-based states for --inject uniform
  --pump-center PUMP_CENTER
                        eV
  --pump-sigma PUMP_SIGMA
                        eV
  --hot HOT             meV above the minimum, for --inject thermal
  --width WIDTH         meV
  --density DENSITY     per-state occupation scale
  --t-end T_END         fs
  --nt NT               number of saved snapshot times, linspace(0, --t-end, --nt) (>= 2). Not the
                        integration steps: the BDF solver picks its own adaptive steps, set by
                        --rtol/--atol
  --workers WORKERS     parallel processes for the out-rate refinement, which dominates the cost
                        at large --refine (per MPI rank under mpirun/srun)
  --threads THREADS     BLAS/OpenMP threads for assembly and propagation on rank 0 (default:
                        OMP_NUM_THREADS as set)
  --checkpoint DIR      save each refined Q in DIR; a rerun skips them
  --mem-per-task GB     memory bound for one refinement task; its triangles are processed in
                        chunks to stay inside it (default XPHD_TASK_MEM_GB or 2)
  --refine-only         stop after the refined out-rates (use with --save-outrates under mpirun)
  --save-outrates FILE
  --outrates FILE       read refined out-rates written by --save-outrates instead of computing
                        them; settings are checked
  --jac {analytic,fd}   Jacobian for the stiff solver. analytic: exact and O(n^2); fd: finite
                        differences, O(n^3), as before 1.1. Same trajectory within --rtol
```

```
  --matrix {dense,sparse,auto}
                        storage for the propagation; sparse pays only when most state pairs are
                        uncoupled
  --rtol RTOL
  --atol ATOL
  --channels            keep channel-resolved rates so the animation can colour each state by the
                        phonon mode filling it. Costs 2*nmodes matrices of nstates^2 in float32.
  --snapshots SNAPSHOTS
```

xphd arpes

```
usage: xphd arpes [-h] [--save SAVE] [--bse-dir BSE_DIR] [--nstates NSTATES]
                  (--temperature TEMPERATURE | --snapshots SNAPSHOTS) [--time TIME] [--emin EMIN]
                  [--emax EMAX] [--ne NE] [--sigma SIGMA] [--valence-shift VALENCE_SHIFT]
                  [--valence-npy VALENCE_NPY] [--qp-db QP_DB] [--path [PATH ...]] [--fine FINE]
                  [--min-occ MIN_OCC] [--lorentz-fwhm LORENTZ_FWHM] [--eta-exph [{full,dilute}]]
                  [--eta-scale ETA_SCALE] [--deg-tol DEG_TOL] [--density DENSITY] [--valence]
                  [--vb-window LO HI] [--vb-ne VB_NE] [--spin-deg SPIN_DEG]
                  [--cell-area CELL_AREA] [--times T [T ...]] [--series FILE] [--no-maps]
                  [--holes POINT [POINT ...]] [--hole-bands HOLE_BANDS]
                  [--hole-radius HOLE_RADIUS] [--hole-frac HOLE_FRAC] [--box POINT RADIUS]
                  [-o OUT]

exciton photoemission I(k, E)

options:
  -h, --help            show this help message and exit
  --save SAVE
  --bse-dir BSE_DIR
  --nstates NSTATES     exciton states per Q
  --temperature TEMPERATURE
                        K; quasi-equilibrium Boltzmann occupations
  --snapshots SNAPSHOTS
                        snaps.npz from `xphd bte`; use with --time
  --time TIME           fs; the snapshot nearest this time is used
  --emin EMIN           eV, on the valence energies' reference (default: automatic)
  --emax EMAX
  --ne NE
  --sigma SIGMA         eV, energy resolution
  --valence-shift VALENCE_SHIFT
                        eV added to the Kohn-Sham valence energies
  --valence-npy VALENCE_NPY
                        full-zone valence energies (nk, nv) in eV, used instead of ns.db1
  --qp-db QP_DB         the GW ndb.QP: quasiparticle corrections added to the Kohn-Sham valence
                        energies, as the GW-BSE exciton energies require
  --path [PATH ...]     high-symmetry labels (G M K K' M' M'' Q Q'), each segment taken the short
                        way: M K G K' M is the straight line through Gamma, as in valley-resolved
                        figures
  --fine FINE           points on the path, evenly spaced in length, with the envelopes and
                        energies interpolated between mesh points; 0 keeps the nearest-mesh-point
                        path only
  --min-occ MIN_OCC     skip excitons with occupation below this fraction of the largest
  --lorentz-fwhm LORENTZ_FWHM
                        eV: convolve every line (excitons and valence band) with a Lorentzian of
                        this FWHM, the line shape of a damping as in yambo; keep --sigma well
                        below it (e.g. 0.005)
  --eta-exph [{full,dilute}]
                        give every exciton line a Lorentzian of its own exciton-phonon linewidth,
                        from the snapshot (xphd bte saves it): 'full' (default) with the
                        stimulated factors at that delay, as Chen, Sangalli and Bernardi (PRR 4,
                        043203) set their eta; 'dilute' the F -> 0 width. On top of --sigma;
                        --lorentz-fwhm adds to it
  --eta-scale ETA_SCALE
                        multiplies the --eta-exph widths (default 1: FWHM = hbar/tau; 2 reads 'eta
                        = Gamma' in 1/(w - E + i eta) as a half width)
  --deg-tol DEG_TOL     eV: with --snapshots, states of one Q closer than this share their
                        occupations (and --eta-exph widths) equally, since the BSE solver's basis
                        inside a degenerate multiplet is arbitrary (0: off)
  --density DENSITY     exciton density, cm^-2, for --valence with --temperature
  --valence             also write the photoemission of the valence band, with the holes the
                        excitons leave taken out of it (needs the number of excitons: the
                        snapshot's own, or --density with --temperature)
  --vb-window LO HI     eV relative to the valence maximum: the energy range of the valence map
                        (default -3 0.5)
  --vb-ne VB_NE         energies of the valence map
  --spin-deg SPIN_DEG   electrons per valence state (2: spin-degenerate)
  --cell-area CELL_AREA
                        nm^2, for --valence with --temperature (default: from the SAVE)
  --times T [T ...]     several delays in one run (with --snapshots): numbers, 'all', or a:b[:s]
                        ranges, fs. -o names one map per delay with {t}, e.g. 77K/arpes_{t}fs.npz
  --series FILE         with --times: the spectra against the delay (k-integrated, in --box,
                        occupied-exciton DOS); default arpes_series.npz beside the maps
  --no-maps             with --times: write the series only
  --holes POINT [POINT ...]
                        also save the replica edge with the hole at these high-symmetry points (G
                        K K' M ...): E(Q) + eps_v(point) - eps_VBM along the path, E(Q) the lowest
                        exciton whose hole sits at the point; plot_arpes.py draws them. Needs
                        --fine > 0
  --hole-bands HOLE_BANDS
                        with --holes: how many valence levels at each point, from the top (2 at K:
                        both spin-split bands)
```

```
  --hole-radius HOLE_RADIUS
                        with --holes: |b| units; the hole must sit this close to the point
  --hole-frac HOLE_FRAC
                        with --holes: the share of the hole weight that must sit there
  --box POINT RADIUS    also integrate the photoemission over a disc of RADIUS (1/Angstrom) about
                        POINT (G, K, M, ...), e.g. --box G 0.12; repeat for several
  -o, --out OUT
```

Parallel runs

xphd hpc-info

```
usage: xphd hpc-info [-h] [--threads THREADS]

options:
  -h, --help         show this help message and exit
  --threads THREADS
```